

JEE MAIN REVISION MASTER: CHEMICAL BONDING AND MOLECULAR STRUCTURE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 26 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	VSEPR Theory & Molecular Geometry	10 PYQs	Molecular Shape
High	Dipole Moment, Formal Charge & Hydrogen Bonding	9 PYQs	Intermolecular
Moderate	Molecular Orbital Theory (MOT) & Bond Order	7 PYQs	MOT

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
VSEPR Theory & Molecular Geometry	$\text{Steric Number } H = \frac{1}{2}(V + M - C + A), \quad \text{LP-LP repulsion} > \text{LP-BP} > \text{BP-BP}$
Dipole Moment, Formal Charge & Hydrogen Bonding	$\mu = q \times d, \quad \text{Formal Charge} = V - L - \frac{1}{2}S, \quad \mu(\text{NH}_3) > \mu(\text{NF}_3)$
Molecular Orbital Theory (MOT) & Bond Order	$\text{Bond Order} = \frac{N_b - N_a}{2}, \quad \text{Stability} \propto \text{Bond Order} \propto \frac{1}{\text{Bond Length}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 VSEPR Theory & Molecular Geometry (High Yield — 10 PYQs)

Core Principle: Prediction of molecular shapes and bond angles using steric number and lone pair repulsions.

■ **Caution / JEE Pitfall:** XeF_4 has sp^3d^2 hybridization with 2 lone pairs; its geometry is octahedral, but its molecular SHAPE is SQUARE PLANAR.

■ **Speed Trick:** In trigonal bipyramidal geometry (sp^3d , e.g. PCl_5), lone pairs ALWAYS occupy equatorial positions to minimize 90° repulsions.

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): "The types of hybrid orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The change in hybridization (if any) of the 'Al' atom in the following reaction is $\text{AlCl}_3 + \text{Cl}^- \rightarrow \text{AlCl}_4^-$..." [Ans: N/A]
- [2021 Shift] Q#1 (MCQ): "Bond enthalpies of A_2 , B_2 and AB are in the ratio 2:1:2. If bond enthalpy of formation of AB is -100kJmol^{-1} . The bond en..." [Ans: N/A]

3.2 Dipole Moment, Formal Charge & Hydrogen Bonding (High Yield — 9 PYQs)

Core Principle: Vector addition of bond dipoles, percentage ionic character, and influence of H-bonding on boiling points.

■ **Caution / JEE Pitfall:** Dipole moment of NH_3 is HIGHER than NF_3 because lone pair and N-H bond moments reinforce each other in NH_3 , but oppose in NF_3 .

■ **Speed Trick:** Intermolecular H-bonding increases boiling point and water solubility (e.g. p-nitrophenol), while intramolecular H-bonding decreases boiling point (o-nitrophenol).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "With respect to resonance structures of CO_3^{2-} ion, which of the following statements are correct? (a) All C-O..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In which of the following option/options, the order of arrangement does not agree with the variation of property indicated against it? (a) \$..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "Dimerisation of solute molecules in low dielectric constant solvent is due to..." [Ans: N/A]

3.3 Molecular Orbital Theory (MOT) & Bond Order (Moderate Yield — 7 PYQs)

Core Principle: LCAO principle, molecular orbital filling order for ≤ 14 vs > 14 electrons, bond order calculation, and magnetic character.

■ **Caution / JEE Pitfall:** O_2 has bond order 2 and is PARAMAGNETIC because it has 2 unpaired electrons in degenerate antibonding $\pi^* 2p_x$ and $\pi^* 2p_y$ orbitals.

■ **Speed Trick:** For species with ≤ 14 electrons (B_2 , C_2 , N_2): $\pi 2p_x = \pi 2p_y$ is filled BEFORE $\sigma 2p_z$ due to s-p mixing.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "The electronic configuration of X and Y are given below: $\begin{aligned} &X: 1s^2 2s^2 2p^6 3s^2 3p^3 \\ &Y: 1s^2 2s^2 2p^6 3s^2 \end{aligned}$ & \$..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "Match List-I with List-II and select the correct option: List-I (Molecule / ion) List-II (Bond order) 1. NO 1. NO 1. NO 1. 15 2. ..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "A pair of isoelectronic species having bond order of one is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

With respect to resonance structures of CO_3^{2-} ion, which of the following statements are correct? (a) All C-O bonds in CO_3^{2-} are equivalent (b) There are three resonance structures possible for CO_3^{2-} ion (c) The position of carbon and oxygen should change in every resonance structure (d) The formal charge on carbon atom is -2

- (A) a, b and c
- (B) a and b only
- (C) b and d only
- (D) a, b and d

Correct Answer: None

Solution Analysis: In the carbonate ion, CO_3^{2-} , carbon is at the centre and the three oxygen atoms are attached to it. The ion has three possible resonance structures. In each structure, one C=O double bond and two C-O⁻ single bonds are shown. The double bond can be placed with any one of the three oxygen atoms. So statement (b) is correct. In resonance, the actual structure is not any one of these structures. The actual carbonate ion is a resonance hybrid. Because of this, all three C-O bonds become equivalent. Each C-O bond has

Benchmark Problem 2: KCET 2026 (Q#3)

The types of hybrid orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are

- (A) sp^3 , sp and sp^2
- (B) sp, sp^2 and sp^3
- (C) sp^2 , sp and sp^3
- (D) sp^2 , sp^3 and sp

Correct Answer: None

Solution Analysis: For finding the hybridisation of nitrogen, we count the number of electron domains around nitrogen. Electron domains include: sigma bonds lone pairs For NO_2^+ : Nitrogen is bonded to two oxygen atoms. Structure is: $\text{O}=\text{N}=\text{O}$ Nitrogen has 2 sigma bonds and no lone pair. So, steric number: $2+0=2$ Steric number 2 means linear shape and sp hybridisation. So, nitrogen in NO_2^+ is sp hybridised. For NO_3^- : Nitrogen is bonded to three oxygen atoms. Due to resonance, all three N-O bonds are equivalent. Nitrogen has 3 sigma b

Benchmark Problem 3: KCET 2025 (Q#1)

The electronic configuration of X and Y are given below: X: $1s^2 2s^2 2p^6 3s^2 3p^3$ Y: $1s^2 2s^2 2p^6 3s^2 3p^5$ Which of the following is the correct molecular formula and type of bond formed between X and Y ?

- (A) X_3Y , ionic bond
- (B) X_2Y_3 , coordinate bond
- (C) XY_3 , covalent bond
- (D) X_2Y , covalent bond

Correct Answer: None

Solution Analysis: The electronic configurations of elements X and Y are as follows: X: $1s^2 2s^2 2p^6 3s^2 3p^3$ Y: $1s^2 2s^2 2p^6 3s^2 3p^5$ For element X, the configuration corresponds to phosphorus (P), which is in group 15 of the periodic table. For element Y, the configuration corresponds to chlorine (Cl), which is in group 17. Phosphorus (P) has a valency of 3 or 5. Chlorine (Cl) has a valency of 1. Given these valencies, possible compounds could form: PCl_3 , where phosphorus uses its valency of 3 to bond with three chlorine atoms. PCl_5 , where phosphorus uses an ex